



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0905 JAVA PROGRAMMING – CO4 – AT2

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JAVA APPLLET LIFE CYCLE

```
import java.applet.Applet;

public class LifeCycle extends Applet {

    public void init() {
        System.out.println("Applet Initialized");
    }

    public void start() {
        System.out.println("Applet Started");
    }

    public void stop() {
        System.out.println("Applet Stopped");
    }

    public void destroy() {
        System.out.println("Applet Destroyed");
    }

}
```

HTML CODE:

```
<html>
<body>
    <applet code="LifeCycle.class" width="400" height="300">
        </applet>
</body>
</html>
```

```
Command Prompt - appletviewer lifecycle.html
java version "1.8.0_491"
Java(TM) SE Runtime Environment (build 1.8.0_491-b10)
Java HotSpot(TM) 64-Bit Server VM (build 25.491-b10, mixed mode)

C:\Users\Dell\Downloads\JAVA>javac welcome.java
welcome.java:4: error: class WelcomeApplet is public, should be declared in a file named WelcomeApplet.java
public class WelcomeApplet extends Applet {
      ^
1 error

C:\Users\Dell\Downloads\JAVA>javac welcome.java

C:\Users\Dell\Downloads\JAVA>appletviewer welcome.html
Warning: Can't read AppletViewer properties file: C:\Users\Dell\.hotjava\properties Using defaults.

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C:\Users\Dell\Downloads\JAVA>appletviewer welcome.html

C:\Users\Dell\Downloads\JAVA>cd "C:\Users\Dell\Downloads\JAVA"

C:\Users\Dell\Downloads\JAVA>javac lifecycle.java

C:\Users\Dell\Downloads\JAVA>appletviewer lifecycle.html
init() called
start() called

Applet Viewer: lifecycle.cl...
Applet

Java Applet Life Cycle

Applet starts
```

```
Command Prompt
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C:\Users\Dell\Downloads\JAVA>javac lifecycle.java

C:\Users\Dell\Downloads\JAVA>appletviewer lifecycle.html
init() called
start() called
stop() called
destroy() called

C:\Users\Dell\Downloads\JAVA>
```

GRAPHICS – SHAPES

```
import java.applet.Applet;
import java.awt.*;

public class GraphicsDemo extends Applet {
    public void paint(Graphics g) {
        g.setColor(Color.CYAN);
        g.fillRect(0, 0, 500, 300);
        g.setColor(Color.YELLOW);
        g.fillOval(400, 30, 60, 60);
        g.setColor(Color.WHITE);
        g.fillOval(60, 40, 70, 40);
        g.fillOval(100, 25, 80, 50);
        g.fillOval(145, 40, 70, 40);
        g.setColor(Color.GREEN);
        g.fillRect(0, 200, 500, 100);
        g.setColor(Color.DARK_GRAY);
        g.fillRect(0, 240, 500, 60);
        g.setColor(Color.WHITE);
        g.fillRect(40, 267, 60, 5);
        g.fillRect(150, 267, 60, 5);
        g.fillRect(260, 267, 60, 5);
        g.fillRect(370, 267, 60, 5);
        g.setColor(Color.ORANGE);
        g.fillRect(180, 130, 140, 90);
        g.setColor(Color.RED);
        Polygon roof = new Polygon();
        roof.addPoint(160, 130);
        roof.addPoint(250, 70);
        roof.addPoint(340, 130);
        g.fillPolygon(roof);
        g.setColor(Color.GRAY);
        g.fillRect(235, 175, 30, 45);
    }
}
```

```

        g.setColor(Color.CYAN);
        g.fillRect(195, 150, 30, 25);
        g.fillRect(275, 150, 30, 25);
        g.setColor(new Color(139, 69, 19));
        g.fillRect(50, 150, 20, 70);
        g.setColor(Color.GREEN);
        g.fillOval(25, 110, 70, 70);
        g.setColor(new Color(139, 69, 19));
        g.fillRect(390, 150, 20, 70);
        g.setColor(Color.GREEN);
        g.fillOval(365, 110, 70, 70);
    }
}

```

HTML CODE:

```

<html>
<body>
<applet code="GraphicsDemo.class" width="500" height="300">
</applet>
</body>
</html>

```

The screenshot shows a Windows Command Prompt window with the following commands and output:

```

C:\Users\Karanya>cd /d "C:\Users\Karanya\OneDrive\Desktop\jdk"
C:\Users\Karanya\OneDrive\Desktop\jdk>javac GraphicsDemo.java
GraphicsDemo.java:48: error: cannot find symbol
    g.setColor(Color.BROWN);
                  ^
    symbol:   variable BROWN
    location: class Color
GraphicsDemo.java:54: error: cannot find symbol
    g.setColor(Color.BROWN);
                  ^
    symbol:   variable BROWN
    location: class Color
2 errors

C:\Users\Karanya\OneDrive\Desktop\jdk>appletviewer GraphicsDemo.html
load: class GraphicsDemo.class not found.
java.lang.ClassNotFoundException: GraphicsDemo.class
    at sun.applet.AppletClassLoader.findClass(AppletClassLoader.java:219)
    at java.lang.ClassLoader.loadClass(ClassLoader.java:424)
    at sun.applet.AppletClassLoader.loadClass(AppletClassLoader.java:152)
    at java.lang.ClassLoader.loadClass(ClassLoader.java:357)
    at sun.applet.AppletClassLoader.loadCode(AppletClassLoader.java:634)
    at sun.applet.AppletPanel.createApplet(AppletPanel.java:799)
    at sun.applet.AppletPanel.runLoader(AppletPanel.java:728)
    at sun.applet.AppletPanel.run(AppletPanel.java:378)
    at java.lang.Thread.run(Thread.java:748)

C:\Users\Karanya\OneDrive\Desktop\jdk>javac GraphicsDemo.java
C:\Users\Karanya\OneDrive\Desktop\jdk>appletviewer GraphicsDemo.html

```

Simultaneously, an Applet Viewer window titled "Applet Viewer: GraphicsDem..." is open, displaying a simple 2D landscape scene. The scene features a blue sky with a white cloud and a yellow sun, a green ground with two green trees, and a central orange house with a red roof and a grey door. The Applet Viewer status bar at the bottom indicates "Applet started".

CALCULATOR

```
import java.applet.Applet;
import java.awt.Graphics;

public class Calculator extends Applet {
    double num1, num2;
    double addition, subtraction, multiplication, division;

    public void init() {
        num1 = Double.parseDouble(getParameter("num1"));
        num2 = Double.parseDouble(getParameter("num2"));
        addition = num1 + num2;
        subtraction = num1 - num2;
        multiplication = num1 * num2;
        division = num1 / num2;
    }

    public void paint(Graphics g) {
        g.drawString("CALCULATOR", 120, 40);
        g.drawString("Number 1 : " + num1, 80, 80);
        g.drawString("Number 2 : " + num2, 80, 110);
        g.drawString("Addition      : " + addition, 80, 150);
        g.drawString("Subtraction   : " + subtraction, 80, 180);
        g.drawString("Multiplication : " + multiplication, 80, 210);
        g.drawString("Division      : " + division, 80, 240);
    }
}
```

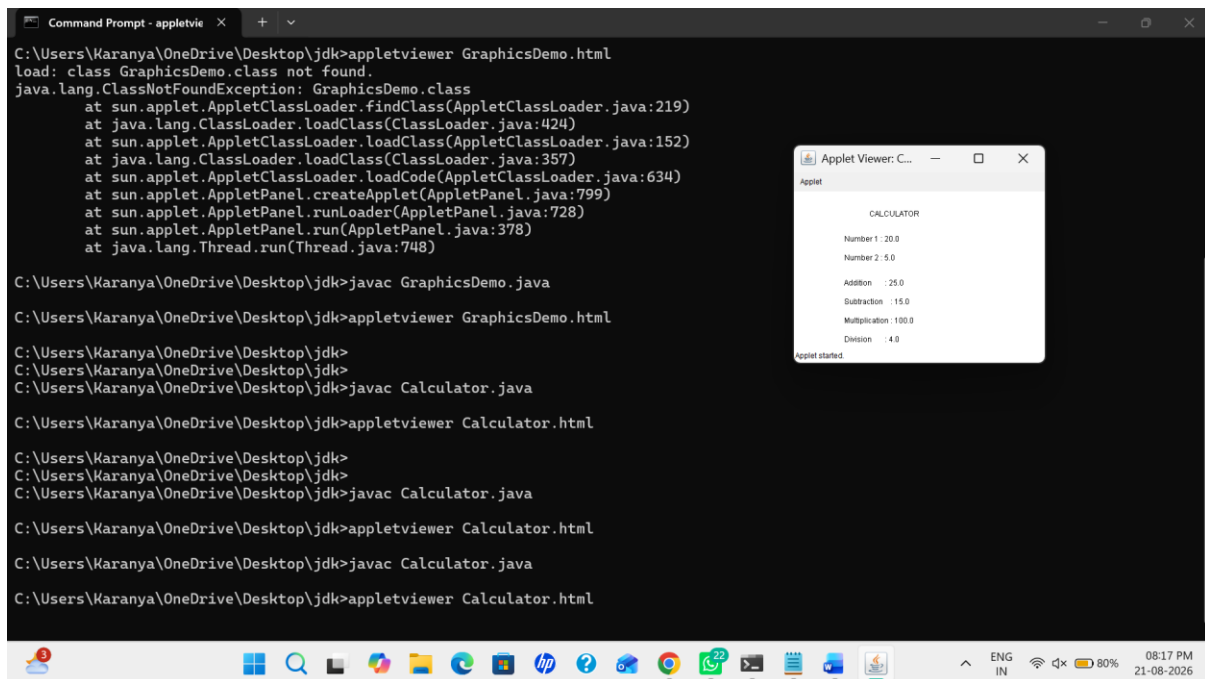
HTML CODE:

```
<html>
<body>
<applet code="Calculator.class" width="400" height="250">
    <param name="num1" value="20">
    <param name="num2" value="5">
    <param name="operator" value="+">
```

</applet>

</body>

</html>



STUDENT REGISTRATION FORM – USING AWT

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
class StudentProfile extends Frame implements ActionListener {
```

```
    TextField nameField, regField;
```

```
    Checkbox male, female;
```

```
    CheckboxGroup genderGroup;
```

```
    Choice department, year;
```

```
    List skills;
```

```
    TextArea report;
```

```
    Button generate, clear;
```

```
    StudentProfile() {
```

```
        setTitle("Student Profile");
```

```
        setSize(600, 600);
```

```
        setLayout(new BorderLayout(10, 10));
```

```
        Label title = new Label("STUDENT PROFILE", Label.CENTER);
```

```
        title.setFont(new Font("Arial", Font.BOLD, 22));
```

```
        add(title, BorderLayout.NORTH);
```

```
        Panel form = new Panel(new GridLayout(7, 2, 10, 10));
```

```
        form.add(new Label("Name:"));
```

```
        nameField = new TextField();
```

```
        form.add(nameField);
```

```
        form.add(new Label("Register Number:"));
```

```
        regField = new TextField();
```

```
        form.add(regField);
```

```
        form.add(new Label("Gender:"));
```

```

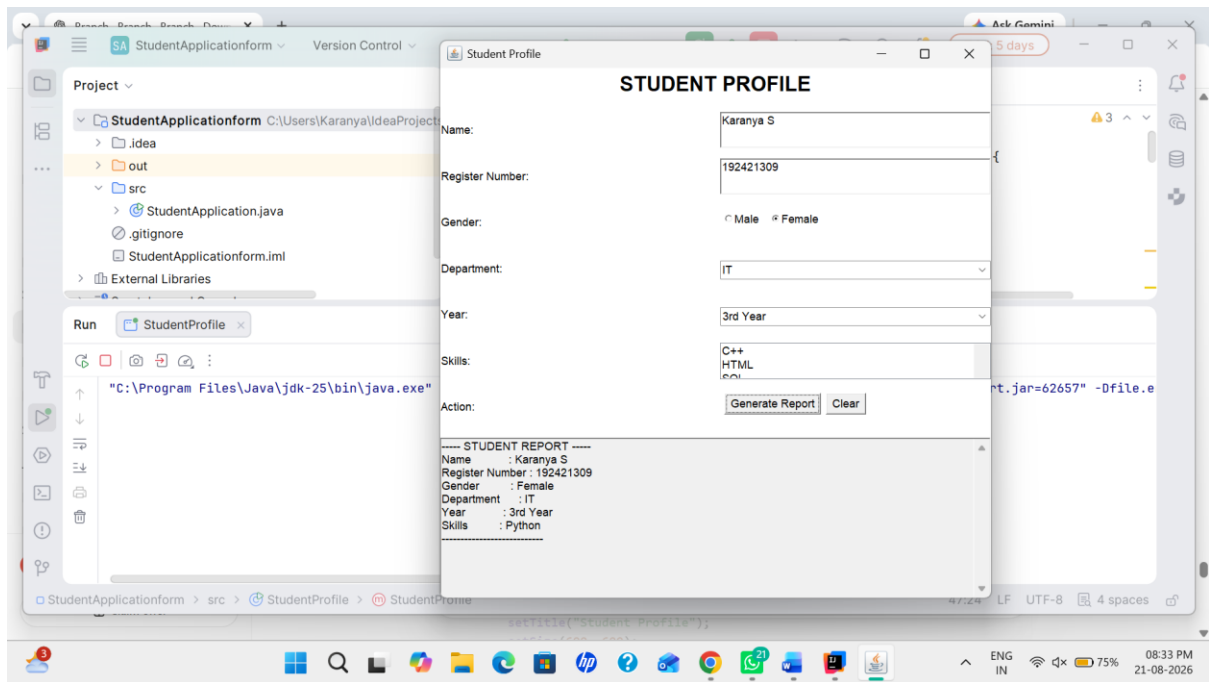
Panel genderPanel = new Panel(new FlowLayout(FlowLayout.LEFT));
genderGroup = new CheckboxGroup();
male = new Checkbox("Male", genderGroup, false);
female = new Checkbox("Female", genderGroup, false);
genderPanel.add(male);
genderPanel.add(female);
form.add(genderPanel);
form.add(new Label("Department:"));
department = new Choice();
department.add("CSE");
department.add("IT");
department.add("ECE");
department.add("EEE");
department.add("MECH");
form.add(department);
form.add(new Label("Year:"));
year = new Choice();
year.add("1st Year");
year.add("2nd Year");
year.add("3rd Year");
year.add("4th Year");
form.add(year);
form.add(new Label("Skills:"));
skills = new List(4, true);
skills.add("Java");
skills.add("Python");
skills.add("C++");
skills.add("HTML");
skills.add("SQL");
form.add(skills);
form.add(new Label("Action:"));
Panel buttonPanel = new Panel(new FlowLayout(FlowLayout.LEFT));
generate = new Button("Generate Report");
clear = new Button("Clear");
generate.addActionListener(this);
clear.addActionListener(this);
buttonPanel.add(generate);
buttonPanel.add(clear);
form.add(buttonPanel);
add(form, BorderLayout.CENTER);
// Report
report = new TextArea();
report.setEditable(false);
add(report, BorderLayout.SOUTH);
// Window closing
addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        System.exit(0);
    }
});

```

```

        setLocationRelativeTo(null);
        setVisible(true);
    }
    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == generate) {
            String gender = "Not Selected";
            if (male.getState())
                gender = "Male";
            else if (female.getState())
                gender = "Female";
            String[] selectedSkills = skills.getSelectedItems();
            String skillText = "";
            for (String skill : selectedSkills) {
                skillText = skillText + skill + " ";
            }
            report.setText(
                "----- STUDENT REPORT -----\\n" +
                "Name          : " + nameField.getText() + "\\n" +
                "Register Number : " + regField.getText() + "\\n" +
                "Gender          : " + gender + "\\n" +
                "Department      : " + department.getSelectedItem() + "\\n" +
                "Year            : " + year.getSelectedItem() + "\\n" +
                "Skills          : " + skillText + "\\n" +
                "-----"
            );
        }
        if (e.getSource() == clear) {
            nameField.setText("");
            regField.setText("");
            report.setText("");
            genderGroup.setSelectedCheckbox(null);
            department.select(0);
            year.select(0);
            skills.deselect(0);
            skills.deselect(1);
            skills.deselect(2);
            skills.deselect(3);
            skills.deselect(4);
        }
    }
    public static void main(String[] args) {
        new StudentProfile();
    }
}

```

LAYOUT IMPLEMENTATION

```
import java.awt.*;
import java.awt.event.*;
class LayoutImplementation extends Frame implements ActionListener {
    CardLayout mainCards;
    Panel mainPanel;
    Button gridButton, gridBagButton, borderButton, cardButton;
    TextField calcDisplay;
    double num1, num2;
    String operator = "";
    TextField studentName, regNo, email;
    Choice department, year;
    Checkbox male, female;
    CheckboxGroup gender;
    CardLayout studentCards;
    Panel studentPanel;
    LayoutImplementation() {
        setTitle("AWT Layout Implementation");
        setSize(700, 600);
        setLayout(new BorderLayout());
        Panel topPanel = new Panel(new FlowLayout());
        gridButton = new Button("GridLayout");
        gridBagButton = new Button("GridBagLayout");
        borderButton = new Button("BorderLayout");
        cardButton = new Button("CardLayout");
        gridButton.addActionListener(this);
        gridBagButton.addActionListener(this);
        borderButton.addActionListener(this);
        cardButton.addActionListener(this);
    }
}
```

```

topPanel.add(gridButton);
topPanel.add(gridBagButton);
topPanel.add(borderButton);
topPanel.add(cardButton);
add(topPanel, BorderLayout.NORTH);
mainCards = new CardLayout();
mainPanel = new Panel(mainCards);
mainPanel.add(createGridLayout(), "grid");
mainPanel.add(createGridBagLayout(), "gridbag");
mainPanel.add(createBorderLayout(), "border");
mainPanel.add(createCardLayout(), "card");
add(mainPanel, BorderLayout.CENTER);
addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        System.exit(0);
    }
});
setLocationRelativeTo(null);
setVisible(true);
}
Panel createGridLayout() {
    Panel main = new Panel(new BorderLayout());
    Label title = new Label("GridLayout Calculator", Label.CENTER);
    title.setFont(new Font("Arial", Font.BOLD, 20));
    main.add(title, BorderLayout.NORTH);
    calcDisplay = new TextField();
    calcDisplay.setEditable(false);
    main.add(calcDisplay, BorderLayout.CENTER);
    Panel buttons = new Panel(new GridLayout(4, 4, 5, 5));
    String[] names = {
        "7", "8", "9", "+",
        "4", "5", "6", "-",
        "1", "2", "3", "*",
        "0", "=", "C", "/"
    };
};
for (String name : names) {
    Button b = new Button(name);
    b.addActionListener(this);
    buttons.add(b);
}
Panel center = new Panel(new BorderLayout());
center.add(calcDisplay, BorderLayout.NORTH);
center.add(buttons, BorderLayout.CENTER);
main.add(center, BorderLayout.CENTER);

return main;
}
Panel createGridBagLayout() {
    Panel panel = new Panel(new GridBagLayout());
    GridBagConstraints gbc = new GridBagConstraints();

```

```

gbc.insets = new Insets(8, 8, 8, 8);
gbc.fill = GridBagConstraints.HORIZONTAL;
Label title = new Label("Student Registration", Label.CENTER);
title.setFont(new Font("Arial", Font.BOLD, 20));
gbc.gridx = 0;
gbc.gridy = 0;
gbc.gridwidth = 2;
panel.add(title, gbc);
gbc.gridwidth = 1;
gbc.gridx = 0;
gbc.gridy = 1;
panel.add(new Label("Name:"), gbc);
studentName = new TextField(20);
gbc.gridx = 1;
panel.add(studentName, gbc);
gbc.gridx = 0;
gbc.gridy = 2;
panel.add(new Label("Register No:"), gbc);
regNo = new TextField(20);
gbc.gridx = 1;
panel.add(regNo, gbc);
gbc.gridx = 0;
gbc.gridy = 3;
panel.add(new Label("Department:"), gbc);
department = new Choice();
department.add("CSE");
department.add("IT");
department.add("ECE");
department.add("EEE");
gbc.gridx = 1;
panel.add(department, gbc);
gbc.gridx = 0;
gbc.gridy = 4;
panel.add(new Label("Email:"), gbc);
email = new TextField(20);
gbc.gridx = 1;
panel.add(email, gbc);
gbc.gridx = 0;
gbc.gridy = 5;
panel.add(new Label("Gender:"), gbc);
gender = new CheckboxGroup();
Panel genderPanel = new Panel(new FlowLayout(FlowLayout.LEFT));
male = new Checkbox("Male", gender, false);
female = new Checkbox("Female", gender, false);
genderPanel.add(male);
genderPanel.add(female);
gbc.gridx = 1;
panel.add(genderPanel, gbc);
gbc.gridx = 0;
gbc.gridy = 6;

```

```

panel.add(new Label("Year:"), gbc);
year = new Choice();
year.add("1st Year");
year.add("2nd Year");
year.add("3rd Year");
year.add("4th Year");
gbc.gridx = 1;
panel.add(year, gbc);
Button submit = new Button("Submit");
Button reset = new Button("Reset");
submit.addActionListener(this);
reset.addActionListener(this);
Panel buttonPanel = new Panel();
buttonPanel.add(submit);
buttonPanel.add(reset);
gbc.gridx = 0;
gbc.gridy = 7;
gbc.gridwidth = 2;
panel.add(buttonPanel, gbc);
return panel;
}
Panel createBorderLayout() {
    Panel main = new Panel(new BorderLayout());
    Label header = new Label("BorderLayout Application", Label.CENTER);
    header.setFont(new Font("Arial", Font.BOLD, 20));
    main.add(header, BorderLayout.NORTH);
    Panel south = new Panel();
    Button home = new Button("Home");
    Button about = new Button("About");
    Button exit = new Button("Exit");
    home.addActionListener(this);
    about.addActionListener(this);
    exit.addActionListener(this);
    south.add(home);
    south.add(about);
    south.add(exit);
    main.add(south, BorderLayout.SOUTH);
    Panel west = new Panel(new GridLayout(3, 1));
    Button menu1 = new Button("Menu 1");
    Button menu2 = new Button("Menu 2");
    Button menu3 = new Button("Menu 3");
    menu1.addActionListener(this);
    menu2.addActionListener(this);
    menu3.addActionListener(this);
    west.add(menu1);
    west.add(menu2);
    west.add(menu3);
    main.add(west, BorderLayout.WEST);
    Panel east = new Panel(new GridLayout(3, 1));
    Button option1 = new Button("Option 1");

```

```

    Button option2 = new Button("Option 2");
    Button option3 = new Button("Option 3");
    option1.addActionListener(this);
    option2.addActionListener(this);
    option3.addActionListener(this);
    east.add(option1);
    east.add(option2);
    east.add(option3);
    main.add(east, BorderLayout.EAST);
    Label center = new Label("Click a button", Label.CENTER);
    center.setFont(new Font("Arial", Font.PLAIN, 18));
    main.add(center, BorderLayout.CENTER);
    return main;
}

Panel createCardLayout() {
    Panel main = new Panel(new BorderLayout());
    Label title = new Label("CardLayout - Student Application", Label.CENTER);
    title.setFont(new Font("Arial", Font.BOLD, 20));
    main.add(title, BorderLayout.NORTH);
    studentCards = new CardLayout();
    studentPanel = new Panel(studentCards);
    Panel card1 = new Panel();
    card1.add(new Label("Student Information"));
    card1.add(new TextField(20));
    studentPanel.add(card1, "card1");
    Panel card2 = new Panel();
    card2.add(new Label("Academic Information"));
    card2.add(new TextField(20));
    studentPanel.add(card2, "card2");
    Panel card3 = new Panel();
    card3.add(new Label("Confirmation / Report"));
    card3.add(new TextArea(5, 25));
    studentPanel.add(card3, "card3");
    main.add(studentPanel, BorderLayout.CENTER);
    Panel buttons = new Panel();
    Button previous = new Button("Previous");
    Button next = new Button("Next");
    Button submit = new Button("Submit");
    previous.addActionListener(this);
    next.addActionListener(this);
    submit.addActionListener(this);
    buttons.add(previous);
    buttons.add(next);
    buttons.add(submit);
    main.add(buttons, BorderLayout.SOUTH);
    return main;
}

public void actionPerformed(ActionEvent e) {
    String command = e.getActionCommand();
    if (command.equals("GridLayout")) {

```

```

        mainCards.show(mainPanel, "grid");
    }
    else if (command.equals("GridBagLayout")) {
        mainCards.show(mainPanel, "gridbag");
    }
    else if (command.equals("BorderLayout")) {
        mainCards.show(mainPanel, "border");
    }
    else if (command.equals("CardLayout")) {
        mainCards.show(mainPanel, "card");
    }
    else if (command.matches("[0-9]")) {
        calcDisplay.setText(calcDisplay.getText() + command);
    }
    else if (command.equals("+") ||
             command.equals("-") ||
             command.equals("*") ||
             command.equals("/")) {
        num1 = Double.parseDouble(calcDisplay.getText());
        operator = command;
        calcDisplay.setText("");
    }
    else if (command.equals("=")) {
        num2 = Double.parseDouble(calcDisplay.getText());
        double result = 0;
        if (operator.equals("+"))
            result = num1 + num2;
        else if (operator.equals("-"))
            result = num1 - num2;
        else if (operator.equals("*"))
            result = num1 * num2;
        else if (operator.equals("/"))
            result = num1 / num2;
        calcDisplay.setText(String.valueOf(result));
    }
    else if (command.equals("C")) {
        calcDisplay.setText("");
    }
    else if (command.equals("Submit")) {
        if (studentName != null) {
            System.out.println("----- STUDENT DETAILS -----");
            System.out.println("Name      : " + studentName.getText());
            System.out.println("Register No: " + regNo.getText());
            System.out.println("Department : " + department.getSelectedItem());
            System.out.println("Email      : " + email.getText());
            System.out.println("Year       : " + year.getSelectedItem());
            System.out.println("-----");
        }
    }
    else if (command.equals("Reset")) {

```

```

        studentName.setText("");
        regNo.setText("");
        email.setText("");
        department.select(0);
        year.select(0);
        gender.setSelectedCheckbox(null);
    }
    else if (command.equals("Exit")) {
        System.exit(0);
    }
    else if (command.equals("Next")) {
        studentCards.next(studentPanel);
    }
    else if (command.equals("Previous")) {
        studentCards.previous(studentPanel);
    }
}
public static void main(String[] args) {
    new LayoutImplementation();
}
}

```

